

County health rankings – Part 1

Problems:

1. Identify the average violent crime rate by state, then break it down further by county to analyse local variations within each state, excluding national-level data and non-state districts such as Washington, D.C.
2. Which counties within each state have the highest violent crime rates?

Solutions:

1. Personal notes: using measure, states, counties and health_statistics table.

```
Query Query History
1 SELECT
2     S.state_full_name,
3     C.county_name,
4     M.measure_name,
5     ROUND(AVG(HS.raw_value)::NUMERIC, 2) AS avg_crime_rate
6 FROM health_statistics HS
7 JOIN states S using(state_id)
8 JOIN Counties C USING (County_id)
9 RIGHT JOIN measures M USING (measure_id)
10 WHERE
11     M.measure_name = 'Violent crime rate'
12     AND S.state NOT IN ('US', 'DC') -- Exclude national average and Washington, D.C.
13     AND HS.raw_value IS NOT NULL
14 GROUP BY
15     S.State_code,
16     S.state_full_name,
17     C.county_name,
18     M.measure_name
19
20 ORDER BY
21     S.state_full_name asc,
22     avg_crime_rate desc;
23
```

(i)Solution Syntax

Data Output Messages Graph Visualiser x Notifications				
	state_full_name	county_name	measure_name	avg_crime_rate
	character varying (100)	character varying (100)	character varying (100)	numeric
62	Alabama	Etowah County	Violent crime rate	139.34
63	Alabama	Lawrence County	Violent crime rate	159.23
64	Alabama	Shelby County	Violent crime rate	157.56
65	Alabama	Cherokee County	Violent crime rate	149.60
66	Alabama	Limestone County	Violent crime rate	89.98
67	Alabama	Choctaw County	Violent crime rate	74.46
68	Alabama	Lamar County	Violent crime rate	42.34
69	Alaska	Dillingham Census Area	Violent crime rate	1847.27
70	Alaska	North Slope Borough	Violent crime rate	922.52
71	Alaska	Anchorage Borough	Violent crime rate	846.03
72	Alaska	Bristol Bay Borough	Violent crime rate	804.48
Total rows: 1000 of 3090 Query complete 00:00:00.202 Ln 23, Col 1				

(ii)Solution Preview

Explanation: US and DC are not included as they are not US states, we have also excluded Null data. Raw_data has been round and converted in numeric data from INT, we have joined 4 tables i.e., fact table – health_statistics, states, counties and measure

2. Personal notes: using same tables as used in (1) and using “DISTINCT ON”

```

Query Query History
1 SELECT DISTINCT ON (S.state_full_name)
2     S.state_full_name,
3     C.county_name,
4     M.measure_name,
5     ROUND(AVG(HS.raw_value)::NUMERIC, 2) AS avg_crime_rate
6 FROM health_statistics HS
7 JOIN states S USING (state_id)
8 JOIN counties C USING (county_id)
9 JOIN measures M USING (measure_id)
10 WHERE
11     M.measure_name = 'Violent crime rate'
12     AND S.state NOT IN ('US', 'DC')
13     AND HS.raw_value IS NOT NULL
14 GROUP BY
15     S.state_full_name,
16     C.county_name,
17     M.measure_name
18 ORDER BY
19     S.state_full_name ASC,
20     AVG(HS.raw_value) DESC;

```

(i)Solution Syntax

	state_full_name character varying (100)	county_name character varying (100)	measure_name character varying (100)	avg_crime_rate numeric
1	Alabama	Chilton County	Violent crime rate	948.40
2	Alaska	Dillingham Census Area	Violent crime rate	1847.27
3	Arizona	Graham County	Violent crime rate	1071.84
4	Arkansas	Crittenden County	Violent crime rate	1479.32
5	California	Alpine County	Violent crime rate	983.37
6	Colorado	Denver County	Violent crime rate	662.37
7	Connecticut	New Haven County	Violent crime rate	377.36
8	Delaware	New Castle County	Violent crime rate	702.25
9	Florida	Putnam County	Violent crime rate	1101.87
10	Georgia	Burke County	Violent crime rate	1155.88
11	Hawaii	Kauai County	Violent crime rate	318.73
12	Idaho	Valley County	Violent crime rate	375.89
13	Illinois	Alexander County	Violent crime rate	1306.94
14	Indiana	Marion County	Violent crime rate	1043.20
Total rows: 50 of 50 Query complete 00:00:00.278 Ln 20, Col 28				

(ii)Solution preview

1. **Explanation:** Using “DISTINCT ON” so that only one row is taken for state and descending order on avg_crime_rate , make sure the highest crime rate corresponding county is shown.